

Methodology Report: E-Commerce Conversion Prediction

Summer Analytics 2026 Hackathon

1. Overview & Strategy

The challenge required binary classification of user conversion (Converted). With a historical conversion prior of ~30.5%, we prioritized precision-recall balance and robustness against covariate shift. Our pipeline was engineered to maximize the F1 score without relying on artificial probability calibration.

2. Feature Engineering & Preprocessing

To maximize information density while preventing target leakage, we concatenated training and public test sets for unsupervised global feature extraction.

- **Intent Metrics:** Derived `Time_per_Page` and `Products_per_Page` to quantify user intent intensity.
- **Synchronized Encoding:** To avoid the Curse of Dimensionality inherent in One-Hot Encoding, we implemented a locked `OrdinalEncoder`. This established a global categorical-to-integer dictionary, ensuring test-set features remained synchronized with training logic. Unseen test categories were routed to an isolated bucket (-1) to prevent inference errors.
- **Missingness:** Missing value indicators were engineered to allow the model to learn the pattern of "bounced" sessions independently of numerical imputation.

3. Validation & Modeling

- **Validation:** We employed Stratified 5-Fold Cross-Validation, ensuring the 30.5% conversion prior was strictly respected within every fold to reduce variance in Out-Of-Fold (OOF) scores.
- **Algorithm Stack:** We utilized LightGBM, XGBoost, and CatBoost. For CatBoost, early stopping was optimized against AUC rather than F1 to prevent premature training convergence caused by rigid probability thresholding.
- **Ensemble Optimization:** We bypassed lazy averaging in Favor of an exhaustive Grid Search across the model simplex. The optimal blend was identified as **65% CatBoost and 35% XGBoost**, which maximized the global F1 objective.

4. Thresholding & Covariate Shift Guardrail

Standard models default to a 0.5 probability threshold, which severely suppresses recall in imbalanced datasets. We performed an exhaustive sweep across the OOF probability space, identifying 0.28 as the global maximum for the F1 metric.